Experiment 2.1

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1. **Aim:** Implement Data link Layer Protocol such as CSMA,CSMA/CD etc.

2. **Objective:** The objective is to implement Data Link Layer protocols, such as CSMA and CSMA/CD, to manage efficient data transmission over shared communication channels by minimizing collisions and optimizing network performance.

3. **Requirements(Hardware/Software):** Cisco Packet Tracer

4. **Procedure:**
   a. Attach required devices (Hub/Switch) in the packet tracer software.
   b. Assign IP address to devices.
   c. Select source and destination and drop packet from source to destination.
   d. Go to Simulation mode and click capture/ play.
   e. When two devices try to send data simultaneously, a collision occurs, which Packet Tracer will show.

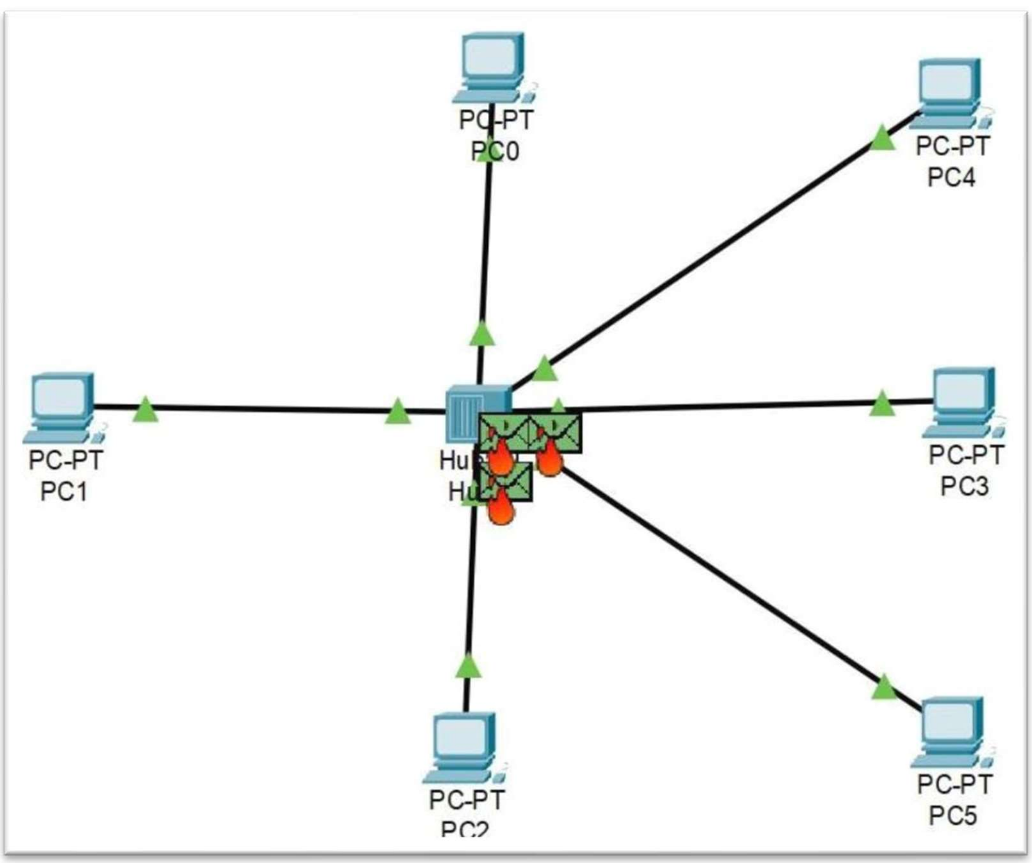
5. **Theory:**

CSMA is a network protocol that controls access to the transmission medium by ensuring that devices check whether the medium is free before attempting to send data. This "carrier sensing" helps to reduce the chances of collisions, which occur when two or more devices transmit simultaneously.

Key Concepts in CSMA:

- **Carrier Sensing:** Before sending data, a device listens to the communication medium to detect if it is busy or free. If the medium is busy, the device waits for it to become free.
- **Random Backoff:** If the medium is busy, the device waits for a random amount of time before checking again. This random delay reduces the likelihood that two devices will attempt to transmit simultaneously after the medium becomes free.
- **Collision Detection:** Devices monitor the network while transmitting data. If a collision is detected (i.e., two devices transmit simultaneously), the devices immediately stop transmitting.
- **Jamming Signal:** After detecting a collision, devices send a jamming signal to ensure all other devices are aware of the collision.

6. **Output:**



7. Learning Outcomes:

- **Understanding Protocol Mechanics:** Gain a deep understanding of how CSMA, CSMA/CD, and other data link layer protocols operate, including how they handle collisions and manage network traffic.
- **Simulation and Implementation Skills:** Develop the ability to simulate and implement these protocols in a programming environment, which involves coding, debugging, and testing.
- **Network Performance Analysis:** Learn to analyze the performance of these protocols under different network conditions, including high and low traffic scenarios, and understand their strengths and limitations.
- **Problem-Solving in Network Design:** Improve problem-solving skills by tackling challenges that arise in network design, such as minimizing collisions and optimizing data flow.
- **Hands-on Experience with Networking Concepts:** Gain practical experience with key networking concepts, preparing for more advanced topics in network design and analysis.
- **Comparison and Evaluation of Protocols:** Develop the ability to compare different data link layer protocols, evaluating their effectiveness and suitability for various network environments